

TE 458 — Network Planning & Optimization: Formula Sheet

Every formula from Modules 1–4 and Tutorial 1 (+ solutions), with explanations and usage notes

All formulas below are transcribed directly from the text layer of the four TE 458 module decks and the Tutorial 1 problem sheet / worked solutions (*TE458_Module1–4.pdf*, *TE 458 TUTORIAL 1.pdf*, *TE458_Tutorial1_Solutions.pdf*, *SOLUTION (2).pdf*) — not re-derived or guessed. Each entry cites the source module/page so you can cross-check against your own copy.

Module 1 — Introduction to Network Planning & Optimization

Module 1 is conceptual only (QoS vs QoE, KPIs, SLAs, CAPEX vs OPEX, planning-vs-optimization trade-offs). It contains no numeric formulas to extract.

Module 2 — Traffic Engineering & Network Dimensioning

Traffic intensity (the Erlang)

$$A = \lambda h$$

A = offered traffic intensity in Erlangs · λ = average arrival rate of calls/sessions · h = average holding time

Use when: converting a call-arrival-rate + average-duration description into an Erlang traffic figure — the starting point for any Erlang B/C dimensioning problem. 1 Erlang = one resource occupied continuously for the whole observation period.

Source: Module 2, p.7–8

Erlang B formula (blocking / loss system)

$$B(N, A) = (A^N / N!) \div \sum_{k=0}^N (A^k / k!)$$

$B(N, A)$ = blocking probability · N = number of circuits/channels/resources · A = offered traffic in Erlangs

Use when: the system has a fixed number of resources and blocked users simply abandon the attempt (no queue) — e.g. trunk circuits, circuit-switched voice channels, legacy cellular voice capacity. Solve for the smallest N such that $B(N, A) \leq$ the target Grade of Service (e.g. 2%). Worked example in the deck: $A=20$ Erlangs, target 2% blocking → $N=28$ channels.

Source: Module 2, p.15, 29–30

Erlang C formula (delay / queueing system)

$$C(N, A) = [(A^N / N!) \cdot (N / (N - A))] \div [\sum_{k=0}^{N-1} (A^k / k!) + (A^N / N!) \cdot (N / (N - A))]$$

$C(N, A)$ = probability an arriving request has to wait · N = number of servers/resources · A = offered traffic in Erlangs (valid only for $A < N$)

Use when: arriving requests wait in a queue instead of being blocked — e.g. call centres, packet-processing systems, shared server pools. Test increasing N until $C(N, A) \leq$ target delay probability.

Source: Module 2, p.18, 32

Module 3 — Propagation Modelling & Coverage Design

EIRP (Effective Isotropic Radiated Power)

$$\text{EIRP (dBm)} = \text{TX power} + \text{antenna gain} - \text{body/feeder loss}$$

Use when: establishing the transmit side of any link budget (uplink: mobile is TX; downlink: base station is TX). Use body loss for a handset held against the body; use feeder loss for the base-station side.

Source: Module 3, p.16; Tutorial 1 formula sheet

Receiver sensitivity (direct / non-spread-spectrum form)

$$S_i \text{ (dBm)} = 30 + E_b/N_0 \text{ (dB)} + F \text{ (dB)} + 10\log_{10}(kT_0) \text{ (dB)} + 10\log_{10}(W) \text{ (dB)}$$

k = Boltzmann's constant (1.38×10^{-23} J/K) · T_0 = 290 K reference temperature · F = receiver noise figure · W = detection bandwidth (Hz) · equivalently S_i = noise floor + E_b/N_0 , where noise floor = $N_0 + 10\log_{10}(W) + F$

Use when: there is **no spreading/despreading gain** to account for — i.e. a plain digital link where the required E_b/N_0 (or SNR) applies directly across the channel bandwidth. This is the form used for the 5G NR downlink/uplink tutorial problems (Problems 1 & 2), not the WCDMA/CDMA problems.

Source: Module 3, p.10–11, 14–15; Tutorial 1 Problems 1–2 & solutions

Thermal noise density, thermal noise power, receiver noise floor

$$N_0 \text{ (dBm/Hz)} = 30 + 10\log_{10}(kT_0)$$

$$\text{Thermal noise power (dBm)} = N_0 + 10\log_{10}(W)$$

$$\text{Receiver noise floor (dBm)} = N_0 + 10\log_{10}(W) + NF$$

Use when: building up a receiver noise budget from first principles (Boltzmann's constant and temperature) rather than being given a lump "thermal noise density" figure. All three lines chain together: density → power over bandwidth W → + noise figure = floor.

Source: Module 3, p.10; Tutorial 1 formula sheet; Tutorial 1 Problem 2 solution

Processing gain (spread-spectrum / CDMA)

$$\text{Processing gain (dB)} = 10\log_{10}(M_{\text{cps}} / R_s)$$

M_{cps} = chip rate · R_s = information/bit rate of the service

Use when: the system is a spread-spectrum/CDMA system (WCDMA/UMTS) — despreading at the receiver "relaxes" the raw SNR requirement by this amount. Higher service bit rate → lower processing gain (they move inversely, since M_{cps} is fixed for the air interface).

Source: Module 3, p.18; Tutorial 1 formula sheet

Total noise + interference

$$\text{Total noise interference (dBm)} = \text{Receiver noise power} + \text{Interference margin}$$

Use when: the problem gives an explicit interference margin (representing other-user/other-cell interference loading, e.g. "7 dB due to 70% cell loading") to add on top of thermal receiver noise, before computing effective sensitivity.

Source: Module 3, p.18; Tutorial 1 formula sheet

Effective receiver sensitivity (CDMA/WCDMA uplink)

Without a stated user count:

$$R_x \text{ (dBm)} = \text{Total noise+interference} - \text{Processing gain} + E_b/N_0$$

With N simultaneously active users stated:

$$R_x \text{ (dBm)} = 10\log_{10}(N) + \text{Total noise+interference} - \text{Processing gain} + E_b/N_0$$

N = number of simultaneously active users

Use when: computing WCDMA/CDMA uplink effective sensitivity. **Critical rule:** include the $10\log_{10}(N)$ term only when the problem explicitly gives a user count (e.g. "considering 20 users"); omit it if no user count is stated. This is the single most commonly missed term in these problems — it is a genuine part of the formula, not optional embellishment, whenever N is given.

Source: Module 3, p.18 (defines N term); Tutorial 1 Solutions p.4, 11–12 (explicit rule + worked contrast between Problem 3 [N given, term included] and Problem 4 [N not given, term omitted])

General link budget (received power)

$$P_r = P_t + G_t + G_r - L_p - L_{\text{misc}}$$

P_r = received power · P_t = transmit power · G_t, G_r = TX/RX antenna gains · L_p = propagation path loss · L_{misc} = other losses (cable, connector, body, penetration)

Use when: deriving or explaining the conceptual link-budget equation. Coverage exists where $P_r \geq P_{\text{sens}}$ (receiver sensitivity).

Source: Module 3, p.21, 23

Maximum Allowable Path Loss (MAPL) — general form

$$L_{p,\text{max}} = P_t + G_t + G_r - L_{\text{misc}} - P_{\text{sens}}$$

Use when: deriving MAPL conceptually from the general link-budget equation (rearranged so $P_r = P_{\text{sens}}$, the coverage-edge condition).

Source: Module 3, p.24

MAPL — practical WCDMA/tutorial "master" form

$$\text{MAPL} = \text{EIRP} - R_x/S_i - (\text{all losses \& margins}) + (\text{all gains})$$

Use when: solving an actual worked problem — this is the template every Tutorial 1 problem plugs into, but which specific losses/margins/gains appear depends on what the question gives you. Only include a term if the problem states it (e.g. Problem 4 gives no base-station antenna gain or feeder loss, so those terms simply don't appear in its MAPL line). Common terms seen across the tutorial: shadow-fading margin, interference margin, fast-fading margin, log-normal fade margin, building penetration loss, base-station antenna gain, feeder/connector loss, soft-handover gain, body loss.

Source: Tutorial 1 formula sheet, p.1; applied in Problems 1–4 solutions

Free-space path loss (Hz & metres form)

$$L_{\text{FS}} \text{ (dB)} = 20\log_{10}(f) + 20\log_{10}(d) - 147.56$$

f = frequency in Hz · d = distance in metres

Use when: f and d are given/converted to Hz and metres respectively. Frequently used at a short reference distance (e.g. $d_0=100$ m) as the anchor for a log-distance model.

Source: Module 3, p.30; Tutorial 1 formula sheet

Free-space path loss (MHz & km form)

$$L_{FS} \text{ (dB)} = 32.44 + 20\log_{10}(f_{\text{MHz}}) + 20\log_{10}(d_{\text{km}})$$

f_{MHz} = frequency in MHz · d_{km} = distance in km

Common pitfall (flagged explicitly in Tutorial 1 Solutions): the two free-space forms use different constants (147.56 vs 32.44) and are **only** equivalent if you keep the matching unit system (Hz/m vs MHz/km). Do not mix a MHz frequency into the 147.56 formula or vice versa.

Source: Module 3, p.25, 32; Tutorial 1 Solutions p.3

Okumura-Hata model (urban macrocell)

$$L_{50} \text{ (dB)} = 69.55 + 26.16\log_{10}(f) - 13.82\log_{10}(h_b) - a(h_m) + [44.9 - 6.55\log_{10}(h_b)]\log_{10}(d)$$

f = frequency in MHz · h_b = base-station antenna height (m) · h_m = mobile antenna height (m) · d = distance in km · $a(h_m)$ = mobile-antenna correction factor

Use when: macrocell coverage planning roughly in the 150–1500 MHz range, across urban/suburban/rural environments. Gives median path loss, not instantaneous fading. In several tutorial problems, plugging in typical macrocell heights ($h_b \approx 30$ m, $h_m \approx 1.5$ m) at a UMTS-band frequency collapses this to the compact form $A + B \cdot \log_{10}(R)$ the exam questions often hand you directly (e.g. $137.4 + 35.2\log_{10}(R)$).

Source: Module 3, p.35–36; Tutorial 1 Solutions p.3, 13

COST-231 Hata model (extends Hata to 1.5–2 GHz)

$$L \text{ (dB)} = 46.3 + 33.9\log_{10}(f) - 13.82\log_{10}(h_b) - a(h_m) + [44.9 - 6.55\log_{10}(h_b)]\log_{10}(d) + C$$

$C = 0$ dB for medium-sized cities/suburban areas · $C = 3$ dB for metropolitan/dense-urban areas

Use when: urban/suburban cellular planning above roughly 1.5 GHz, or whenever the question specifically names "COST-231" / "dense urban" (e.g. Tutorial 1 Problem 4 uses a COST-231 dense-urban variant, $PL_{\text{max}} = 137.8 + 35.4\log_{10}(R)$).

Source: Module 3, p.37; Tutorial 1 Solutions p.3, 13

Mobile-height correction factor $a(h_m)$

$$a(h_m) = (1.1\log_{10}f - 0.7)h_m - (1.56\log_{10}f - 0.8)$$

Use when: you need to actually evaluate $a(h_m)$ rather than being told it's negligible. In the tutorial's worked example this term came out to ≈ 0.05 dB (negligible) for typical UMTS-band parameters — don't assume it's always negligible for other frequency/height combinations.

Source: Tutorial 1 Solutions p.13 (medium-city COST-231 correction form)

Log-distance path loss model

$$PL(d) = PL(d_0) + 10n \cdot \log_{10}(d/d_0)$$

n = path-loss exponent (environment-dependent) · d_0 = reference distance · $PL(d_0)$ = path loss at the reference distance (often computed via free-space loss)

Use when: the question gives a path-loss exponent n and a reference distance/loss rather than a named empirical model (Okumura-Hata/COST-231) — e.g. Tutorial 1 Problem 2 uses $n=3.5$, $d_0=100$ m.

Source: Module 3, p.38; Tutorial 1 formula sheet; Problem 2 solution

Log-normal shadowing model

$$PL(d) = PL(d_0) + 10n \cdot \log_{10}(d/d_0) + X_\sigma$$

X_σ = zero-mean Gaussian random variable (shadowing effect, standard deviation σ)

Use when: the log-distance model needs an added random shadowing term for statistical/probabilistic coverage analysis (as opposed to a single deterministic radius).

Source: Module 3, p.38

Coverage area of a hexagonal cell

$$A = (3\sqrt{3}/2) \cdot R^2 \approx 2.598 \cdot R^2$$

R = cell radius

Use when: converting a computed cell radius into a coverage area, for hexagonal-cell tessellation assumptions (used throughout Tutorial 1 for area/percentage-change questions).

Source: Tutorial 1 formula sheet, p.1; used in Problems 2 & 4 solutions

Module 4 — Access Network Capacity & Interference Planning

Cell capacity (simplified)

$$C_{\text{cell}} = B \cdot \eta$$

B = available bandwidth · η = spectral efficiency (bits/s/Hz)

Use when: making a first-order capacity estimate from bandwidth and an assumed/average spectral efficiency figure.

Source: Module 4, p.6, 28

Average user throughput

$$R_{\text{user}} \approx C_{\text{cell}} / N_{\text{active}}$$

N_{active} = number of active users sharing the cell

Use when: estimating a simple per-user share of total cell capacity. Explicitly flagged in the deck as a simplification — real throughput also depends on SINR, scheduling, mobility and traffic type.

Source: Module 4, p.7

Cluster size (hexagonal reuse)

$$N = i^2 + ij + j^2$$

i, j = non-negative integers · typical values $N = 1, 3, 4, 7, 9, 12, \dots$

Use when: determining valid cluster sizes for a hexagonal frequency-reuse layout. Smaller $N \rightarrow$ more frequent reuse, higher capacity, more interference risk.

Source: Module 4, p.11

Frequency reuse factor

$$\text{Reuse factor} = 1/N$$

Use when: expressing what fraction of total channels each cell gets in a cluster of size N . (Some textbooks call N itself the "reuse factor" — the deck notes this naming inconsistency explicitly.)

Reuse distance

$$D = R\sqrt{3N}$$

D = distance between cells reusing the same frequency group · R = cell radius · N = cluster size

Use when: estimating how far apart co-channel cells must be in a hexagonal layout. Worked example: R=1 km, N=7 → D=√21≈4.58 km.

Source: Module 4, p.12, 15

Channels per cell / total system channels

$$S = kN \quad \Leftrightarrow \quad k = S/N$$

S = total available channels in the system · N = cluster size · k = channels assigned per cell

Use when: splitting a total channel budget S (e.g. from S = total bandwidth ÷ channel bandwidth) across N cells in a cluster. Worked GSM example: 50 MHz ÷ 200 kHz = 250 channels total; with N=4, k=62.5 channels/cell.

Source: Module 4, p.13, 17–18

System capacity (multiple clusters)

$$C = M \cdot k \cdot N = M \cdot S$$

M = number of clusters in the system

Use when: scaling from one cluster's capacity to the full system's capacity across M repeated clusters. Worked example: 7 clusters × 4 cells/cluster × 500 calls/cell (8 calls/channel × 62.5 channels/cell) = 14,000 simultaneous calls.

Source: Module 4, p.14, 19–20

SINR (Signal-to-Interference-plus-Noise Ratio)

$$\text{SINR} = S / (I + N) \quad \text{SINR}_{\text{dB}} = 10 \log_{10} (S / (I + N))$$

S = desired signal power · I = interference power · N = noise power

Use when: assessing signal *quality* (not just strength) — determines achievable modulation/coding scheme and throughput. Coverage (RSRP) alone is not sufficient; a covered user can still have poor SINR.

Source: Module 4, p.24

Spectral efficiency

$$\eta = R / B$$

R = data rate · B = bandwidth · η in bits/s/Hz

Use when: quantifying how efficiently a system converts bandwidth into throughput; feeds directly into the C_{cell}=B·η capacity estimate.

Source: Module 4, p.26

Shannon capacity (ideal channel capacity)

$$C = B \cdot \log_2 (1 + \text{SINR})$$

SINR here in linear scale (not dB)

Use when: estimating a theoretical upper-bound capacity from bandwidth and SINR. Note the logarithmic

(diminishing) return from improving SINR versus the linear return from adding bandwidth.

Source: Module 4, p.27

5G NR subcarrier spacing

$$\Delta f = 2^{\mu} \times 15 \text{ kHz}, \quad \mu = 0, 1, 2, 3, 4$$

Use when: reasoning about 5G NR numerology, in contrast with LTE's fixed 15 kHz subcarrier spacing.

Source: Module 4, p.33

Quick-reference: which sensitivity/MAPL form for which system?

System type	Receiver sensitivity form	MAPL form
5G NR (plain SNR, no spreading)	$S_i = \text{noise floor} + E_b/N_0$	MAPL = EIRP – S_i – margins (+ any stated gains)
WCDMA/UMTS (CDMA, no user count given)	$R_x = \text{Total noise+interference} - \text{Processing gain} + E_b/N_0$	MAPL = EIRP – R_x – margins + gains
WCDMA/UMTS (CDMA, N users stated)	$R_x = 10\log_{10}(N) + \text{Total noise+interference} - \text{Processing gain} + E_b/N_0$	MAPL = EIRP – R_x – margins + gains

Always build MAPL from only the terms the specific question actually gives you — do not assume every margin/gain in the "master" MAPL line applies to every problem.